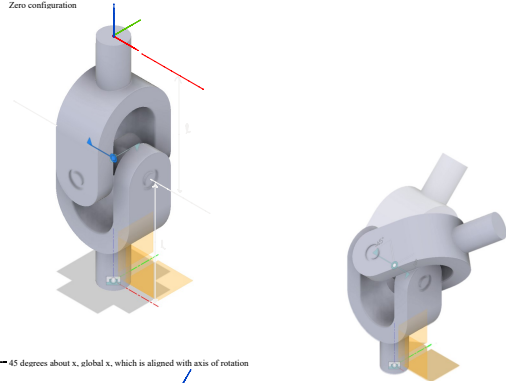


What are the forward kinematics of a universal joint?

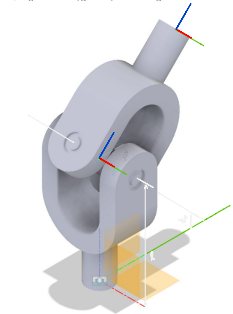
need to figure this one out and then test what happens as I add more universal joints and the length becomes close to 0. Should approximate curvature. But I still don't know if sum of bending about each is the same.

I know for sure the kinematics of fixed length universal joint that has the same tip orientation cannot match that of the curvature.

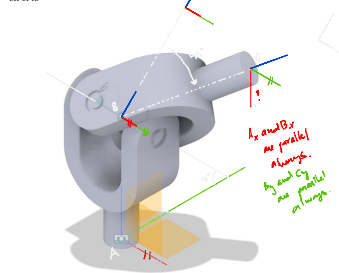
Zero configuration



→ 45 degrees about x, global x, which is aligned with axis of rotation



there's 45 degrees about body fixed y axis, which is $\pi/4$ along aligned with global y axis. It's 45 degrees off of it.



For sure the kinematics of a single universal joint do not approximate those of this continuum. But as the lengths of the links become smaller and smaller, does it? How can I show that?

By algebra

$${}^A P_{B/A} = {}^A P_{B/C} {}^C P_{C/B}$$

$$\begin{aligned} {}^A P_{B/A} &= [0, 0, l]^T \\ {}^A P_{C/B} &= {}^A P_{B/C} {}^C P_{C/B} \\ &= {}^A R_B {}^B R_C [0, 0, l]^T \\ &= R_A(\theta_1) R_B(\theta_2) [0, 0, l]^T \end{aligned}$$

$${}^A P_{C/A} = \begin{bmatrix} 0 \\ 0 \\ l \end{bmatrix}^T \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta_1 & -s\theta_1 \\ 0 & s\theta_1 & c\theta_1 \end{bmatrix} \begin{bmatrix} c\theta_2 & 0 & s\theta_2 \\ 0 & 1 & 0 \\ -s\theta_2 & 0 & c\theta_2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ l \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ l \end{bmatrix} + \begin{bmatrix} c\theta_2 & 0 & s\theta_2 \\ s\theta_2 c\theta_1 & c\theta_1 & -s\theta_1 c\theta_2 \\ -s\theta_2 c\theta_1 & s\theta_1 & c\theta_1 c\theta_2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ l \end{bmatrix}$$

now to first

$$= \begin{bmatrix} l s\theta_2 \\ -l s\theta_2 c\theta_1 \\ l + l c\theta_1 c\theta_2 \end{bmatrix}$$

If I make a homogeneous transform and do this, then doing chain, how else can I get to g from Allen?

$$\Rightarrow {}^A g_C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\theta_1 & -s\theta_1 & 0 \\ 0 & s\theta_1 & c\theta_1 & l \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_2 & 0 & s\theta_2 & l s\theta_2 \\ 0 & 1 & 0 & 0 \\ -s\theta_2 & 0 & c\theta_2 & l c\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

match? !!

$$= \begin{bmatrix} c\theta_2 & 0 & s\theta_2 & l s\theta_2 \\ s\theta_2 c\theta_1 & c\theta_1 & -s\theta_1 c\theta_2 & -l s\theta_1 c\theta_2 \\ -s\theta_2 c\theta_1 & s\theta_1 & c\theta_1 c\theta_2 & l c\theta_1 c\theta_2 + l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

can't remember angular velocity approach... I think this is it?

$${}^A W_{C/A} = {}^A W_{C/B} + {}^A W_{B/A}$$

Now if I make $q_1 = q_2$ and $\dot{q}_1 = \dot{q}_2$ and multiply several of these matrices together, but constraining $q_1 = q_2$ and $\dot{q}_1 = \dot{q}_2$ for all matrices (assume each universal joint is delinked same amount)?

10 disks
= 9 universal joints
= l is the gap/2

$$\text{i.e. } ({}^A g_B)^n$$

What happens as $n \rightarrow \infty$? l gets smaller

Does this g look like the g in the Allen paper? symbolically, what is this look like?

garbled mess